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Flare Dynamic Art

GED TRAIL TO ELENKI

SÃO PAULO
2025

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Epigraph

O coração tem razões que a própria
razão desconhece.

Blaise Pascal, Pensées

Resumo

REIS, Arthur Alberto Cardoso; SANTOS, Felipe Gomes Rodrigues dos; TEIXEIRA, Kil Matheus Gomes; CUSTODIO, Leandro de Oliveira. **Trail to Elenki**. 2025. 16 f. TCC (Graduação em **Ciência da Computação, Engenharia de Software e Engenharia da Computação**) – Instituto de Tecnologia e Liderança, São Paulo, 2025.

O presente trabalho apresenta a estruturação da startup FLARE Dynamic Art e o planejamento para seu futuro lançamento conjunto com um jogo autoral independente (*indie*), focado em experiências filosóficas e emocionais. O objetivo central foi consolidar as bases operacionais e estratégicas do empreendimento antes de sua entrada no mercado. O escopo do projeto abrangeu o desenvolvimento técnico do Produto Mínimo Viável (MVP) na *engine* Godot, integrando arte e programação, aliado a uma profunda estruturação de negócios. Foram realizados estudos detalhados de custos, despesas e estimativas tributárias para a análise financeira, bem como a elaboração de estratégias de marketing e de uma proposta de valor voltada para investidores. A validação do modelo de negócios e do produto ocorreu por meio de banca especializada e testes com usuários. Os *feedbacks* obtidos confirmaram a viabilidade do projeto, fundamentando o plano para o lançamento simultâneo da empresa e de seu primeiro título.

Palavras-Chave: Empreendedorismo; Jogos Indie; Análise Financeira; Godot; Plano de Negócios.

ABSTRACT

REIS, Arthur Alberto Cardoso; SANTOS, Felipe Gomes Rodrigues dos; TEIXEIRA, Kil Matheus Gomes; CUSTODIO, Leandro de Oliveira. **Trail to Elenki**. 2025. 16 p. Final Course Project (Bachelor's Degree in Computer Science, Software Engineering, and Computer Engineering) – Institute of Technology and Leadership, São Paulo, 2025.

This work presents the structuring of the startup FLARE Dynamic Art and the planning for its future launch alongside an independent (indie) authorial game, focused on philosophical and emotional experiences. The central objective was to consolidate the operational and strategic bases of the venture before its market entry. The project scope covered the technical development of the Minimum Viable Product (MVP) in the Godot engine, integrating art and programming, combined with a deep business structuring. Detailed studies of costs, expenses, and tax estimates were conducted for financial analysis, as well as the elaboration of marketing strategies and a value proposal aimed at investors. The validation of the business model and the product occurred through a specialized panel and user tests. The feedback obtained confirmed the project's feasibility, grounding the plan for the simultaneous launch of the company and its first title.

Keywords: Entrepreneurship; Indie Games; Financial Analysis; Godot; Business Plan.

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1 INTRODUCTION

This project is part of the Entrepreneurship Track as a graduation requirement, consisting of the simultaneous development of a digital product and a corporate structure. The scope encompasses the creation of the startup **Flare Dynamic Art** and the development of its first original game, **Trail of Alya** (academic title: *Trail to Elenki*).

The work consolidates two main fronts:

1. **Business Structuring:** Complete governance, legal, and financial documentation to ensure the business's viability and its market entry.
2. **Technical Development (MVP):** Creation of a playable Minimum Viable Product, validated through user testing and market research, demonstrating the team's technical and artistic capabilities.

The final objective is to deliver a validated product and a solid business plan, ready for investment rounds and commercial launch.

2 Executive Summary

This document serves as the **Consolidated Public Report**, unifying the development progress from the project's inception through its alpha conceptualization to the final prototype refinement (Modules 13, 14, and 15).

It covers the complete structuring of *Flare Dynamic Art* as a company (governance, legal, and financial foundations) and the full development cycle of the game *Trail of Alya*. The report details the journey from initial lore creation and concept art to the technical implementation of procedural generation, complex AI behaviors (Boss Fight), 3D/2D asset integration, and user validation through playtests.

3 Objectives

The primary objectives achieved during this comprehensive development period include:

- **Company Formation:** Structuring of documentation regarding leadership, governance, legal compliance (LGPD), and financial planning.
- **Game Design:** Creation of a cohesive world lore, storyline, unique language systems, and evolving the Game Design Document (GDD).

- **Technical Engineering:** Implementation of procedural room generation, state-machine-based AI, combat mechanics (combos, weapon switching), and inventory systems.
- **Artistic Production:** Development of a unique visual identity mixing 2D Pixel Art (Tilesets/UI) with 3D Low Poly models (Characters/Bosses), enhanced by custom shaders and lighting.
- **Validation:** Conducting market research, playtests, and iterative bug fixing based on user feedback.

4 Comprehensive Development Report

4.1 Narrative and World Building

- **Lore & Story:** Complete creation of the world lore, character backstories, and the concept for game stages.
- **Immersion:** Development of a unique in-game language system to enhance world depth.

4.2 Art and Visual Assets

- **Concept Art:** Character design, item design, and environmental concepts.
- **2D Assets (Pixel Art):**
 - Creation of extensive Tilesets (Walls, Floors V2/V4/V5, Shelves, Doors).
 - Animated environmental objects (Candles, Totems, Chests).
 - Visual Effects (VFX) for attacks (Seismic waves, projectiles).
- **3D Modeling & Animation:**
 - **Protagonist:** Alya (Modeling, movement animations, static bug fixes).
 - **Enemies:** Hug, Rug, Flamma Monk, and Lost Faces (Modeling and specific attack loops).
 - **Boss:** Full 3D model creation with complex animations (Molten Wax attack, Fireballs).
- **Technical Art (Shaders/VFX):**
 - Implementation of global lighting systems affecting gameplay.
 - "Ghost effect" for dashes and projectiles.
 - Particle systems for flames and magic.

4.3 Engineering and Programming

- **Core Systems:**
 - **Procedural Generation:** Logic using matrices to control room instances, preventing overlaps and dead-end doors.
 - **State Machines:** Dynamic management of Player, Enemy, and Boss behaviors.
- **Combat Mechanics:**

- Implementation of weapon switching (Laser vs. Boomerang) and chained combo systems.
- Defensive mechanics (Dash and Electric Shield).
- Physics adjustments for melee and projectile aiming.
- **Boss Fight Implementation:**
 - Complex AI programming including multiple attack patterns (Dash, Fire rain, Rapid fire).
 - Integration of dialogue systems triggering before combat.
- **UI/UX Programming:**
 - Full HUD implementation (Health bars, Weapon wheel, Item magazines).
 - Menu systems (Main, Pause, Game Over, Settings).
 - Inventory and Key collection logic (4-key system).

4.4 Audio Design

- Composition of original music tracks for gameplay and menus.
- Development and synchronization of sound effects (SFX) for combat and environment.

4.5 Business, Marketing & Documentation

- **Corporate Structure:** Definition of legal foundations, organizational structure, and meeting minute protocols.
- **Identity:** Brand development (Logo, Color Palette, Fonts).
- **Market Strategy:** Market analysis, target audience research, and creation of a Business Plan.
- **Pitch Materials:** Development of Pitch Scripts (EN/PT) and presentation decks for investors.
- **Outreach:** Creation of project website, social media channels, and gameplay trailers.

5 Technologies & Tools Used

- **Game Engine:** Godot Engine 4.3 (Programming in GDScript).
- **IDE:** VSCode.
- **Art & Modeling:** Blockbench, Blender (3D); Aseprite, Piskel, Sketchbook (2D/Concept).
- **Audio:** Reaper (DAW) with Roland-GO:Piano.
- **Project Management:** Google Sheets, Docs, Meeting Minutes.
- **Communication:** Google Meet, Slack.
- **Video/Streaming:** OBS Studio.

6 Challenges and Solutions (Consolidated)

Over the course of the project, the team faced and overcame several key challenges:

Challenge Area	Description	Solution Implemented
Artistic Production	Initial difficulties with specific tools and techniques for 3D/2D integration.	Self-Directed Learning: Utilization of tutorials and internal knowledge sharing sessions to level up team skills.
Communication	Forgetting discussion topics and decisions made during meetings.	Documentation Protocol: Implemented systematic meeting minutes shared among all members to ensure alignment and historical tracking.
Workflow	Improper interference where members altered work areas belonging to others, causing rework.	Validation Pipeline: Adopted a new internal validation method with strict ownership of work areas to prevent unauthorized changes.
Time Management	Accumulation of small tasks and "feature creep" affecting deadlines.	Prioritization Matrix: Adopted the "Priority x Urgency" matrix (suggested by project monitoring) to focus on essential product refinement.

7 Next Steps

With the consolidation of the prototype and documentation, the future roadmap includes:

1. **Final Polish:** Refinement based on the latest Game Feedback and Playtests.
2. **Strategic Re-evaluation:** Updating the market strategy based on the finished prototype.

3. **Investment Rounds:** Conducting the final Pitch presentation for the board and potential investors.
4. **Public Release Strategy:** Planning the roadmap for a potential demo release or early access.

8 Final Considerations

This consolidated report demonstrates the significant maturity achieved by *Flare Dynamic Art*. From a theoretical concept, the team has successfully delivered a functional, aesthetically unique, and technically complex game prototype (*Trail of Alya*).

We have established a solid foundation not only for the game software but also for the company structure. We are now positioned to transition from development to business validation, aiming to attract future investors and drive the product's growth in the market.